

PYTHON FOR BEGINNERS · DAY 7

Lists

Storing many values in one place

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```
fruits = ["apple", "kiwi"]
fruits.append("mango")

for f in fruits:
    print(f)
```

TODAY'S GOALS

By the end of today, you can…

- Create a **list** and read items
- Loop through a list
- Check items with **in**
- Change, add and remove items
- Use **sum**, **max**, **sorted**…
- Build a to-do list app ✅

WHY LISTS?

5 students… or 100?

Without a list 😩

```
student1 = "Ali"
student2 = "Sara"
student3 = "Ahmad"
student4 = "Mariam"
student5 = "Omar"
```

With a list 😎

```
students = ["Ali", "Sara",
    "Ahmad", "Mariam",
    "Omar"]
```

Many values, **one** variable.

CREATING A LIST

Square brackets, commas between

```
scores = [85, 92, 78]
prices = [1.5, 3.25]
mixed  = ["Ali", 20, True]
empty  = []

print(len(scores))  # 3
```

Items keep their **order**

A list can hold **any type** — even mixed

Lists can be **changed** (unlike strings!)

len() counts the items

READING ITEMS

Indexes work just like strings

"Ali"	"Sara"	"Ahmad"	"Mariam"	"Omar"
0	1	2	3	4
-5	-4	-3	-2	-1

`students[0]` → **Ali**

`students[-1]` → **Omar**

`students[1:3]` → **Sara, Ahmad**

CHANGING A LIST 🖍️

Change, add and remove

fruits = ["apple", "kiwi"]	What it does
fruits[1] = "mango"	change an item
fruits.append("plum")	add to the end
fruits.insert(0, "fig")	add at a position
fruits.remove("apple")	remove by value
fruits.pop()	remove the last item (and give it back)
del fruits[0]	remove by index

LOOPS + LISTS 🔁

Go through every item

Code

```
students = ["Ali", "Sara"]

for s in students:
    print("Hello,", s)

if "Sara" in students:
    print("Sara is here!")
```

Output

```
Hello, Ali
Hello, Sara
Sara is here!
```

in checks if an item is in the list.

LIST TOOLS 🧰

scores = [85, 92, 78, 90, 66]

`len(scores)`

5

`sum(scores)`

411

`max(scores)`

92

`min(scores)`

66

`sum / len`

82.2

`sorted(scores)`

66 78 85 90 92

⚠️ WATCH OUT

sorted() vs .sort()

sorted() gives back a **new** list.

.sort() changes the list **itself**.

```
nums = [3, 1, 2]

new = sorted(nums)
print(nums)  # [3, 1, 2]
print(new)   # [1, 2, 3]

nums.sort()
print(nums)  # [1, 2, 3]
```

🛠️ PROJECT

A to-do list app

```
todo = []
while True:
    choice = input("1 Add 2 Show 3 Quit: ")
    if choice == "1":
        todo.append(input("Task: "))
    elif choice == "2":
        for task in todo:
            print("-", task)
    elif choice == "3":
        break
```

This uses **everything** so far:

- a list · a while loop
- if / elif · input()
- a for loop · break

The full version (with “remove”) is in the notebook.

🧠 QUICK CHECK

Predict before you run!

Q1. Which item?

```
x = ["a", "b", "c"]
print(x[-1])
```

Q2. What is printed?

```
n = [1, 2, 3]
n.append(4)
n[0] = 10
print(n)
```

Q3. True or False?

```
pets = ["cat", "dog"]
print("fish" in pets)
```

SUMMARY

What we learned today

Code	Meaning
x = ["a", "b"]	create a list
x[0] · x[-1]	first · last item
x[0] = "z"	change an item
x.append() · x.insert()	add items
x.remove() · x.pop()	remove items
for item in x:	loop through the list
len · sum · min · max · sorted	list tools

💻 YOUR TURN

Practice time!

Open **exercises.ipynb** in the Day 7 folder and solve the 10 tasks, from ⭐ to ⭐ ⭐ ⭐.

⭐ Your favourite movies list

⭐ ⭐ Score statistics until “done”

⭐ ⭐ ⭐ Quiz with lists 🧠

Next lesson: Functions 🧩